SCENTINEL

OBP candidate selection · analysis report

iGEM URV 2025

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Target VOC       :  methyl salicylate
CAS              :  119-36-8
Top N candidates :  10
Interferents     :  caryophyllene, β-Cyclocitral, diisobutyl phthalate, 10-undecanal, 1-octen-3-ol, acet...
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```
tau (selectivity):  5
w1 affinity      :  0.450
w2 selectivity   :  0.250
w4 stability     :  0.150
w5 promiscuity   :  0.150
```

SMAA configuration

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Method 2 (Perturbation):  c = 50%,  100,000 iterations
Method 3 (pure SMAA)   :  100,000 iterations
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SCENTINEL · Detailed results | VOC: methyl salicylate

#	OBP	Conserv.	Conf.	c2	s1	s2	s4	s5	Ki (uM)	Interf	Danger	Type
1	DcitOBP3	0.4130	0.78	0.16	0.472	0.122	1.000	0.902	7.01	7/43	! 6	Classic OBP
2	GmolOBP7	0.3978	0.82	0.28	0.381	0.088	1.000	0.964	9.81	12/43	! 2	Classic OBP
3	AcerOBP11	0.3925	0.77	0.12	0.479	0.082	1.000	0.814	6.83	5/43	! 4	Classic OBP
4	GmolPBP2	0.3855	0.77	0.12	0.452	0.200	0.850	0.793	7.55	5/43	! 2	PBP
5	AcerOBP2	0.3699	0.75	0.07	0.427	0.066	1.000	0.888	8.29	3/43	! 2	Classic OBP
6	AlinCSP1	0.3668	0.82	0.30	0.414	0.200	0.400	0.994	8.67	13/43	! 1	CSP
7	HoblOBP2	0.3645	0.80	0.21	0.248	0.200	1.000	0.970	16.04	9/43	-	Classic OBP
8	RpadOBP3	0.3642	0.74	0.07	0.350	0.162	1.000	0.951	11.00	3/43	! 2	Classic OBP
9	SmosOBP12	0.3611	0.77	0.12	0.386	0.110	1.000	0.780	9.62	5/43	! 4	Classic OBP
10	PxylGOBP2	0.3554	0.79	0.19	0.437	0.025	0.850	0.795	7.97	8/43	! 7	GOBP2

Conserv. = Score x Confidence · c2 = selectivity confidence · Interf = measured/total · Danger = vineyard interferents binding as strong as the target
Row tint: green = confidence >= 0.80 lilac = 0.50-0.80 pink = < 0.50

Dangerous vineyard interferents per candidate

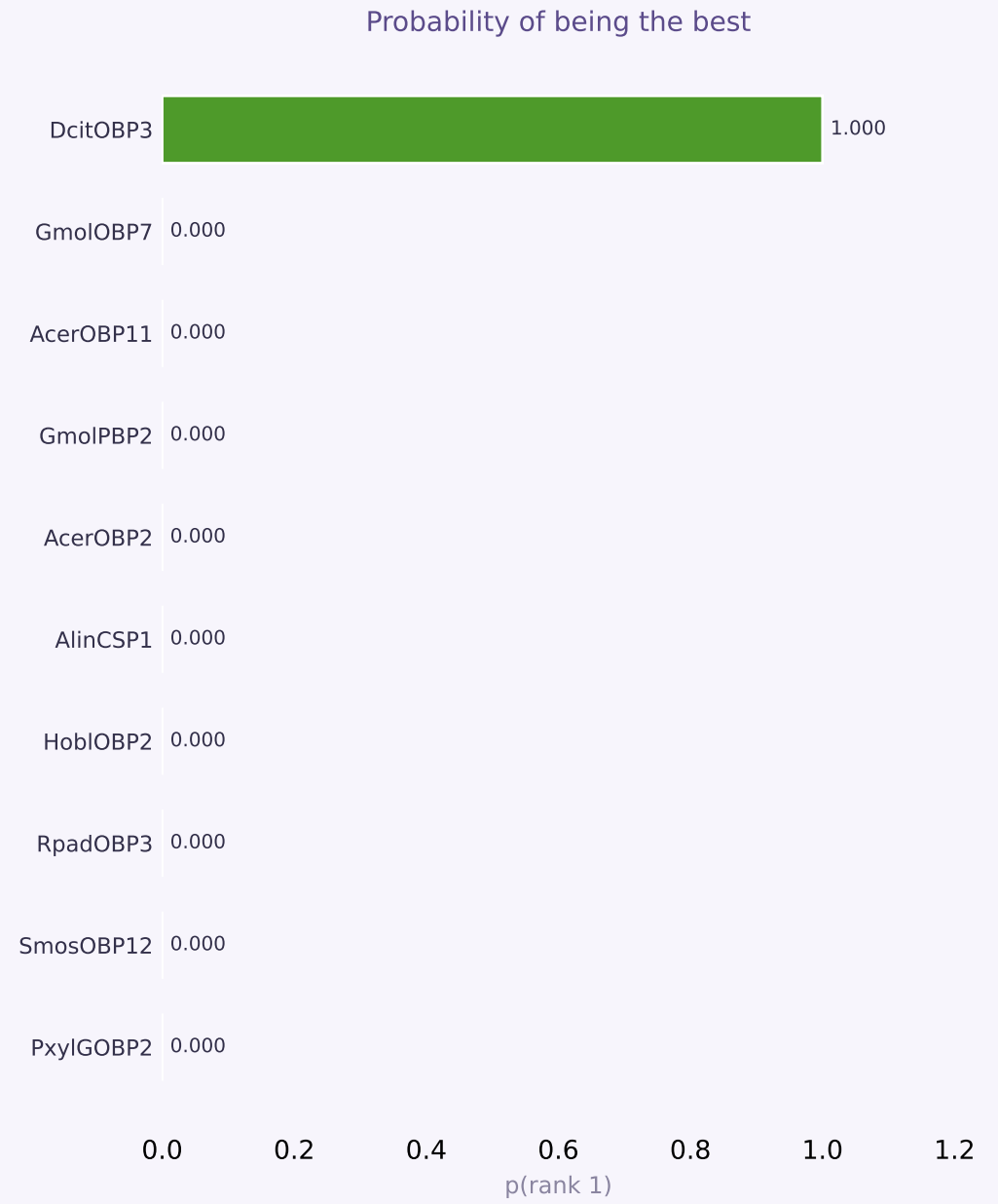
(Ki_interf <= Ki_target x 1.33; v = binds stronger than the target)

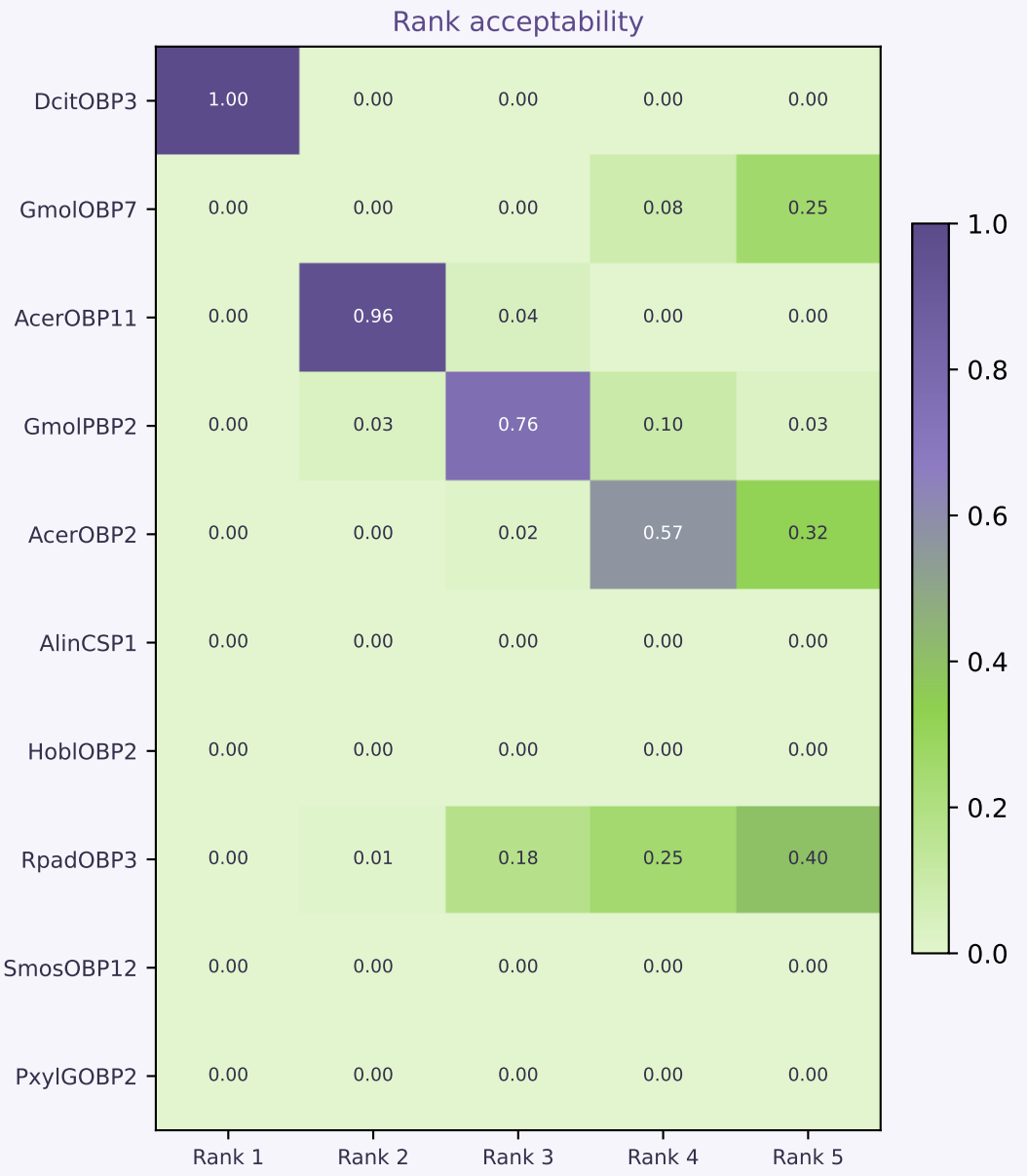
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 1. DcitOBP3         Ki=  7.01 [6]  C9:Ald(4.3v), acetic acid(5.3v), γ-Terpinene(6.8v), r-limonene(8.2), caryophyllen...
 2. GmolOBP7         Ki=  9.81 [2]  12:Ald(4.3v), 10:Ald(12.7)
 3. AcerOBP11        Ki=  6.83 [4]  1-hexanol(2.8v), phenylethanal(3.8v), Ethyl hexadecanoate(4.4v), geraniol(8.6)
 4. GmolPBP2         Ki=  7.55 [2]  C9:Ald(7.6), hexanal(8.0)
 5. AcerOBP2         Ki=  8.29 [2]  phenylethanal(2.7v), geraniol(5.5v)
 6. AlinCSP1         Ki=  8.67 [1]  caryophyllene(9.2)
 8. RpadOBP3         Ki= 11.00 [2]  beta-farnesene(8.9v), r-limonene(10.5v)
 9. SmosOBP12        Ki=  9.62 [4]  1-octen-3-ol(5.3v), 2-ethyl-1-hexanol(6.4v), 12:Ald(8.1v), Ocimene(8.2v)
10. PxylGOBP2        Ki=  7.97 [7]  Farnesene(1.0v), caryophyllene(2.4v), phenylethanal(4.2v), geraniol(5.5v), 1-hexa...
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SCENTINEL · Experimental vs Computational | VOC: methyl salicylate

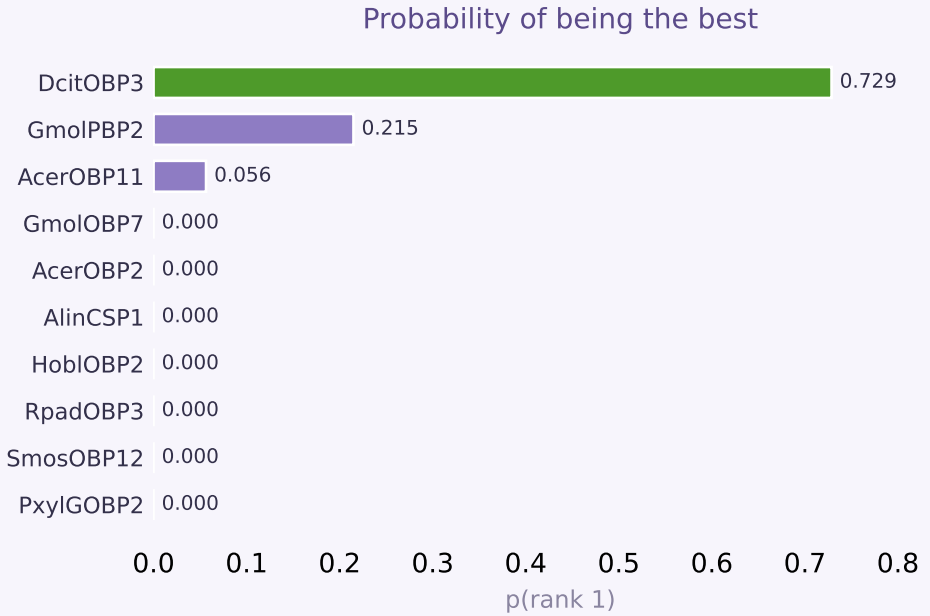


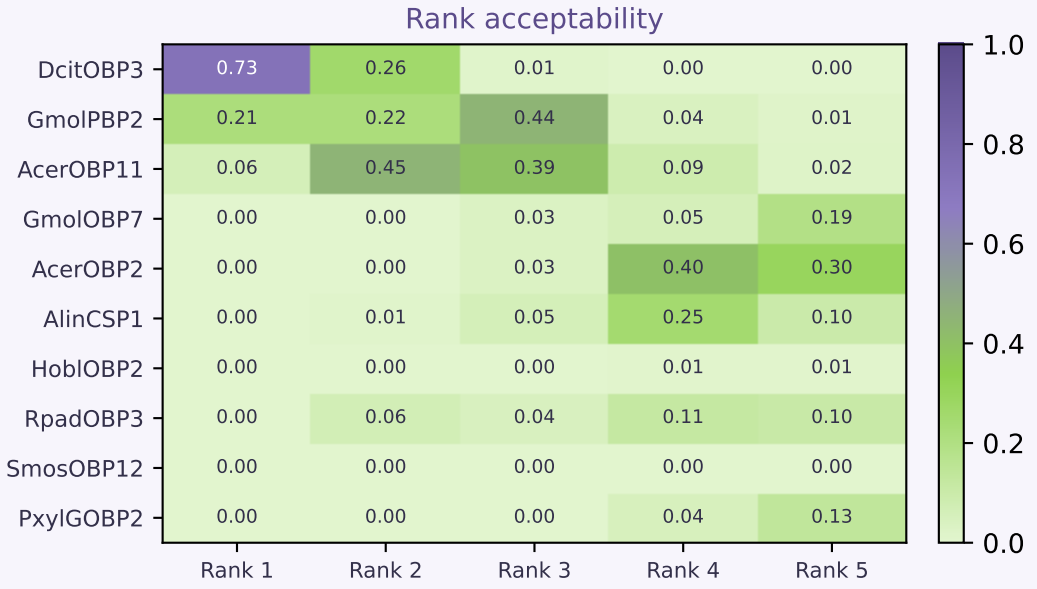
SCENTINEL · Method 2 - Weight Perturbation | VOC: methyl salicylate

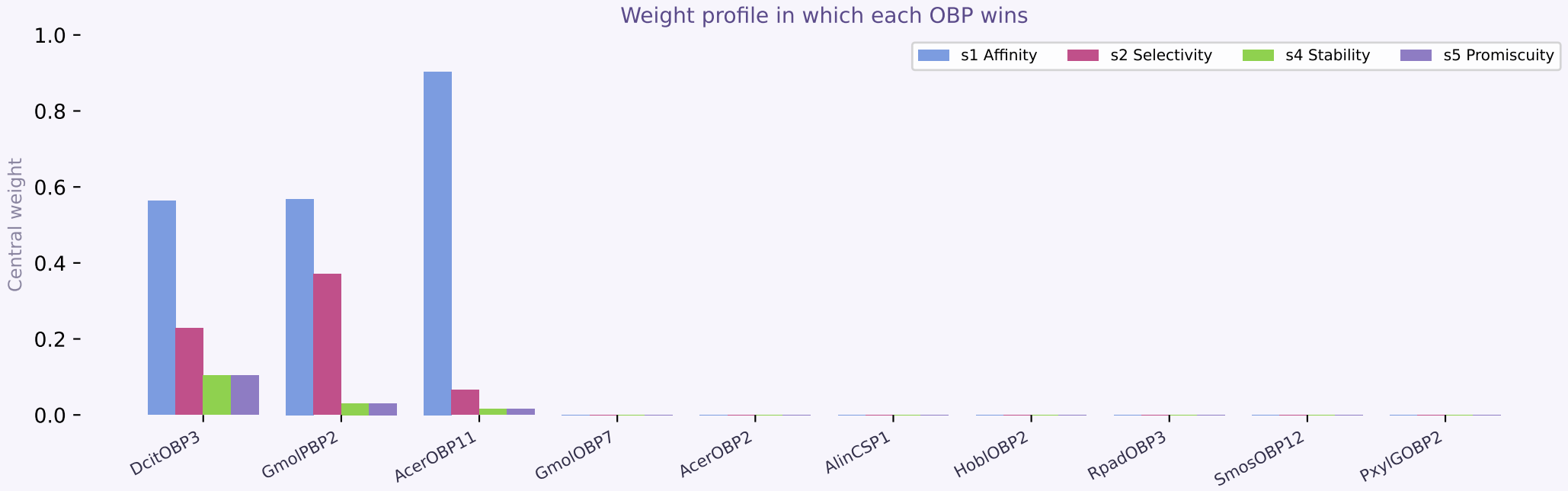




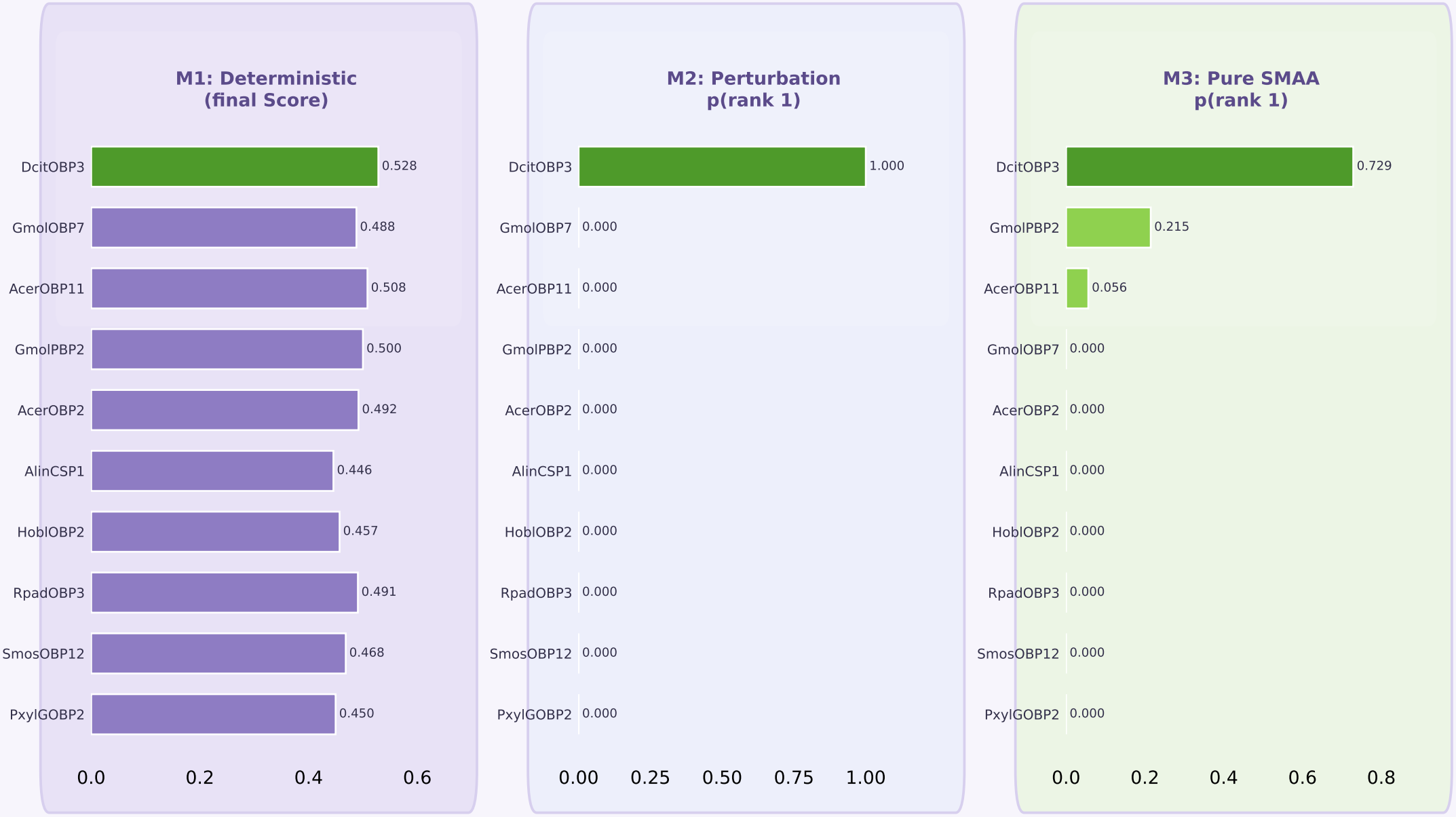
SCENTINEL · Method 3 - Pure SMAA | VOC: methyl salicylate







SCENTINEL · Comparison of the 3 methods | VOC: methyl salicylate



Winner · M1 (deterministic): DcitOBP3 M2 (perturbation): DcitOBP3 M3 (pure SMAA): DcitOBP3